FARHAD SHADMAND

AI Researcher and Developer

I am a researcher and developer with hands-on experience building deep learning models based on diffusion models, GANs, and VLM-based systems on multi-GPU platforms (CUDA). My work has led to multiple publications, including CVPR, WACV, and IEEE Access. I bring strong skills in PyTorch, computer vision, model optimization, and AI agents.

Content

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Farhadsh1992@gmail.com

Citizen of Portugal

https://farhadshad.com/cv/

www.linkedin.com/in/farhadsh1992

www.github.com/farhadsh1992/

Language

English (Fluent)
Farsi (Native)
Portuguese (basic)

Programming

Python, C++, Java (Basic)
AI/ML: **PyTorch, TensorFlow,** ONNX, PyTorch Lightning, **huggingface**, DeepSpeed, langchain, langsmith, langgraphy
Multi-GPU, CUDA,
OpenCV, Pillow, Scikit-learn
SQL, MySQL, Spark, pandas, ChromaDB

Project

- VISUAL-ID – Unique Visual Identities in Graphics, Images and Faces (Link)
- TruIM – Trust Image (Link)
- FACING (link)
- link: https://visteam.isr.uc.pt/team/farhad-shadmand-2/

Entertainment

- Dance (Afro-Latin, Forro)
- Swimming
- Gaming
- Movie

Experience

AI Researcher Internship

Aug-2025 - Nov 2025

Adobe - Paris, France.

- Developed robust watermarking models using geometric symmetry-aware convolutional networks combined with a VLM-based generative noise-simulation pipeline to accurately replicate print-and-scan degradation effects. This approach significantly enhanced watermark resilience against real-world distortions introduced during printing, scanning, and image recapture.
- **Managers:** Dr. Shruti Agarwal (shragarw@adobe.com).

AI Researcher

2019 - 2026

VisTeam - Instituto de Sistemas e Robótica - Coimbra, (ISR)

- Developed deep learning models for steganography, watermarking, object detection, face detection, and face verification. Contributed to the VISUAL-ID, TruIM, and FACING projects, resulting in multiple publications and practical solutions for identity-document security.

Education

PhD. In Electrical Engineering and Intelligent Systems

2021 – 2026

University of Coimbra, Portugal

- My PhD focused on developing printable steganography and **watermarking models** for identity-document security, combining encoder–decoder architectures, geometric-symmetric convolutions, and advanced noise-simulation pipelines. I designed and optimized several deep-learning frameworks for reliable message embedding under real print-and-capture distortions. The work resulted in multiple publications, including IEEE Access, CVPR, and WACV.
- **Supervisors:** Prof. Nuno Gonçalves (nunogon@deec.uc.pt) and Prof. Luiz Schirmer (Luiz.schirmer@ufsm.br)

M.Sc. in Complex system, Physics

2015 - 2017

Isfahan University of Technology, Iran

- I specialize in predicting and modeling stock market behavior, utilizing advanced data analysis and machine learning techniques to forecast trends and inform investment strategies.
- **Supervisor**: Prof. Farhad Shahbazi (shahbazi@iut.ac.ir)

Bachelor in Physics

2011 - 2015

Isfahan University of Technology, Iran

Among the top 10% of graduates.

Innovation patent

- Inventors: Nuno Gonçalves and Farhad Shadmand, Applicants: Imprensa Nacional Casa da Moeda and University of Coimbra. Encoding, decoding and integrity validation systems for a security document with a steganography-encoded image and methods, computer programs and associated computer-readable data carrier. In PT117136A (Portugal), EP4064095A1 (European Patent Office). **Granted in Oct. 2025**.

Papers References

- **StampOne: Addressing Frequency Balance in Printer-proof Steganography - CVPR -2024.**
- **CodeFace: A Deep Learning Printer-Proof Steganography for Face Portraits. IEEE Access 9 (2021)**
- **StylePuncher: Encoding a Hidden QR Code into Images, ICPRAM (2025),**
- **DocSafe: Towards Practical Print-Proof Image steganography via Frequency Decomposition and Covariance Alignment", IEEE Access (2026.3680290), 2026**

- RiemStega: Covariance-based loss for print-proof transmission of data in images, WACV (2025).
- Young Labeled Faces in the Wild (YLFW): A Dataset for Children Faces Recognition - IEEE 18th International Conference on Automatic Face and Gesture Recognition (FG) - (2024)
- Mordeephy: Face morphing detection via fused classification-12th International Conference on Pattern Recognition Application and Methods - (2022)
- Towards facial biometrics for id document validation in mobile devices. - Applied Sciences 11.13 (2021)